

Decent Coaching Classes
IX CBSE
Mind Your Math - IV

Subject: Mathematics Chapters: 2, 3, 4, 5

Marks: 30

Time: 1 hour

Instructions:

- i. This question paper has four sections: A, B, C and D.
- ii. Section A has 5 MCQs of 1 mark each.
- iii. Section B has 4 questions of 2 marks each.
- iv. Section C has 3 questions of 3 marks each. All contain an internal choice as indicated.
- v. Section D has 2 questions of 4 marks each.
- vi. Draw neat, labelled figures/graphs wherever necessary.

Section A – Multiple Choice Questions

(5)

1. Which of the following is an irrational number?
(A) 4 (B) $22/7$ (C) $\sqrt{9}$ (D) $\sqrt{7}$
2. The degree of the polynomial $p(x) = 7$ is:
(A) 0 (B) 1 (C) 7 (D) not defined
3. $a^3 - b^3 =$ _____
(A) $(a - b)(a^2 + ab + b^2)$
(B) $(a + b)(a^2 - ab + b^2)$
(C) $(a - b)(a^2 - ab + b^2)$
(D) $(a - b)^3$
4. The perpendicular drawn from the centre of a circle to a chord _____ the chord.
(A) trisects (B) bisects (C) doubles (D) none of these
5. The rational number equivalent to the repeating decimal $0.777\dots$ is:
(A) $7/9$ (B) $7/10$ (C) $77/100$ (D) $7/99$

Section B – Very Short Answer Questions

(8)

1. Classify the following as rational or irrational: (i) $\sqrt{16}$ (ii) $\sqrt{11}$
2. Express the repeating decimal $0.666\dots$ in the form p/q .
3. Find the value of the polynomial $p(x) = 3x - 5$ at $x = -2$.
4. Using a suitable identity, evaluate: 998^2

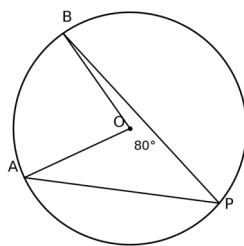
OR

Factorise: $x^2 + 9x + 18$

Section C – Short Answer Questions

(9)

1. Prove that $\sqrt{3}$ is an irrational number.
2. In the figure given below, O is the centre of the circle and the chord AB subtends an angle of 80° at the centre. Find the measure of $\angle APB$, where P is a point on the major arc. Give reason for your answer.



3. Expand using a suitable identity: $(3a + 2b - 1)^2$

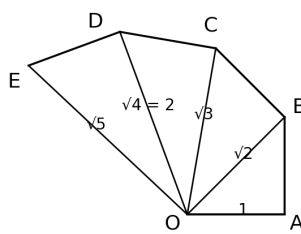
OR

Factorise using a suitable identity: $8p^3 + 27q^3$

Section D – Long Answer Questions

(8)

1. Using the square-root-spiral (Pythagorean spiral) method, construct a length equal to $\sqrt{5}$ units, starting from a unit segment $OA = 1$ unit. Draw the complete spiral up to $OE = \sqrt{5}$ and briefly justify, using the Baudhāyana–Pythagoras theorem, how each successive length ($\sqrt{2}$, $\sqrt{3}$, $\sqrt{4}$, $\sqrt{5}$) is obtained.

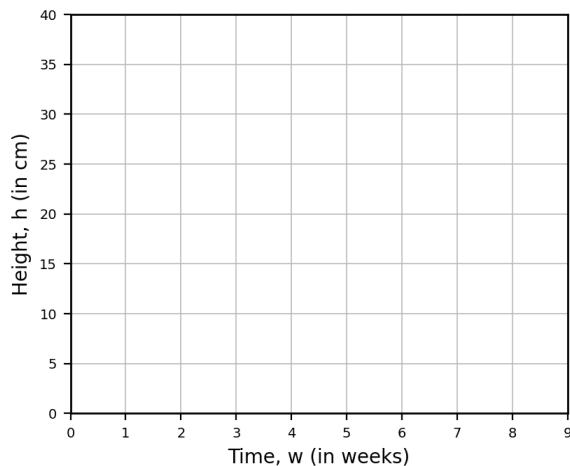


2. A plant is 12 cm tall when planted and grows at a constant rate of 3 cm per week. Let $h(w)$ denote its height (in cm) after w weeks.

(i) Write the linear polynomial $h(w)$ representing the height of the plant.

(ii) Find the height of the plant after 6 weeks.

(iii) On the grid given below, plot the graph of $h(w) = 3w + 12$ for $w = 0, 2, 4, 6$ and 8 . Use your graph to estimate after how many weeks the plant will be 30 cm tall.



— End of Paper —